

Chasing the Nines

Reliable RL Across States and Seeds



Ala Sleimi · Mykhailo Shtandenko
Supervisor: Lennart Röstel · Technical University of Munich

PENDULUM · 2,501 STARTS · FIVE INDEPENDENTLY TRAINED ACTORS
RL + supervised has 9 near-reference failures versus 497 for pure RL; pure RL wins strictly more often.

RL + supervised
9 near-reference failures
Task 93.858% · strict wins 12.555%

Near-reference 99.928%

Pure RL
497 near-reference failures
Task 92.875% · strict wins 22.823%

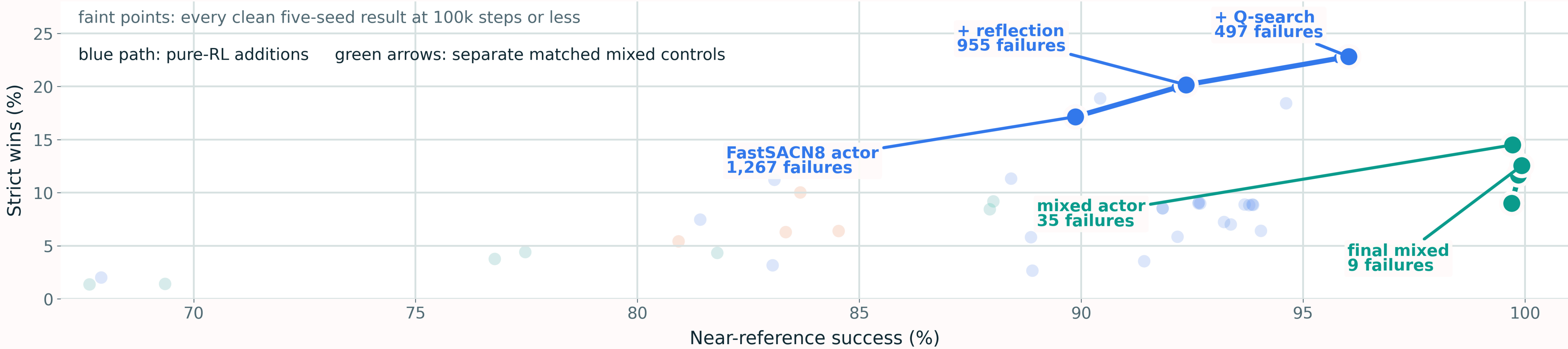
Near-reference 96.026%

01 Results at 100k steps or less

RL + supervised
Learner-state DAgger → priority refit → local Q-search

Pure RL
SimbaV2 FastSACN8 → reflection → 41-action Q-search

Near-reference reliability and strict wins measure different behavior. The blue path improves both. The green controls cluster near 100% reliability.



02 Reliability is a tail problem

497
pure-RL near-reference failures

VS

9
RL + supervised near-reference failures

Five actors × 2,501 starts. Each heatmap cell counts actor seeds below the near-reference threshold.

RL + supervised
9 misses on 8 starts

Pure RL
497 misses on 208 starts

Initial angular velocity [-1, 1]
Initial angle [-180°, 180°]

Number of actor seeds that fail from each start

0	1	2	3	4	5
---	---	---	---	---	---

All five actors pass 2,493 RL + supervised starts and 2,293 pure-RL starts. Pure-RL seeds disagree on 195 starts; all five fail on 13.

Protocol: 61 angles × 41 angular velocities, 200-step rollouts. **Near:** within 5 return of the better controller. **Task:** upright and slow for ≥80%. **Strict:** beats both controllers. Five seeds, 12,505 trials.

03 Inside the pure-RL networks

Matched target ablation inside the pure-RL networks

Matched intervention: same SimbaV2, five seeds, 100k transitions, and one update per transition. Only the critic target changes: SAC is one-step, Fast averages one- and eight-step targets, and N8 is eight-step. Actor tests use 5,553 states per seed; critic tests use 512 failures per seed. Thin lines are seeds; dots are medians.

A Actor bound

B Actor slope

C Critic test

D Clip effect

■ C blue: $\partial Q/\partial a$ points toward the helpful action
■ C teal: Q ranks the helpful action above the actor
■ C orange: the projected critic step lowers return
■ D gray: actor is at the torque bound
■ D blue: $\partial Q/\partial a$ points beyond the bound
■ D orange: fraction of the proposed step retained

$a\theta(s)=2 \tanh u\theta(s), \partial a/\partial u=2[1-\tanh^2(u)] \quad a'=\Pi_{[-2,2]} \{a+0.05 \operatorname{sign}(\partial Q/\partial a)\}$

Actor bottleneck: Fast and N8 raise torque-bound occupancy from 59% to 87% and reduce median output sensitivity from 0.120 to 0.039. On roughly 75% of failures, $\partial Q/\partial a$ points toward a higher-return controller action, yet only about 7% of the proposed step survives clipping. Scoring 41 torques directly avoids that local actor projection and raises reflection from 94.5% to 95.9%.

04 What changes the tail?

Matched intervention minus control: net classifications on 5 × 2,501 paired trials

Only component changed	Near	Task	Strict
Learner-state refit vs no refit	+29	+7	+442
Priority vs uniform sampling	+10	+2	+111
Mixed Q-search vs same actor	+26	+32	-243
Reflection vs same pure actor	+312	+10	+376
Pure Q-search vs reflection	+458	+242	+333

Joint-loss gradient scale

A Norm ratio

B Cosine

37.2×
median $\|\beta_{BC}\| / \|\beta_{SAC}\|$

-0.943 to +0.935
gradient cosine range

One randomly initialized SimbaV2 SACN8 seed, 25k steps, static reference anchors: joint BC plus SACN8 reaches 2,443 near-reference starts; reward-only SACN8 reaches 1,757. These are actor gradients. One coefficient was tested; normalized or projected joint losses remain open.

RL plus supervision leaves 9 near-reference failures in 12,505 trials. In matched pure-RL runs, multistep actors retain about 7% of a local critic step after clipping; 41-action Q-search improves reflection from 94.5% to 95.9%.

References: dynamic programming and energy controller
SAC: Haarnoja et al., 2018 · DAgger: Ross et al., 2011
SimbaV2: Lee et al., 2025 · SACn: Łyskawa et al., 2025